

Peak-Hour Delay Recovery

Reducing food-delivery cancellations through better diagnosis, honest communication, and economically justified compensation.

Project type: Independent product strategy exercise | Context: Swiggy/Zomato-style marketplace (not affiliated)

My contribution: Defined the investigation, evaluated possible interventions, proposed customer targeting, and developed an experiment and unit-economics framework.

Decision in one sentence: Prioritize clear delay communication as a baseline; pilot targeted compensation only where incremental customer value is likely to exceed incentive costs.

Evidence standard

This is a scenario-based case study. Operational findings, experimental outcomes, and financial inputs below are illustrative assumptions, not observed company data. No real experiment was run, and no customer outcome is claimed.

What this case demonstrates

- Root-cause thinking: trace cancellations from timing to order lifecycle and marketplace capacity.
- Prioritization: distinguish a root-cause fix from a short-term service-recovery measure.
- Business judgment: reject blanket discounts unless unit economics support them.

01 / Define and diagnose the problem

Problem: Customers cancel orders after experiencing uncertain or extended delivery times, especially during the dinner rush.

Business objective: Reduce avoidable cancellations without harming contribution margin or trust.

Diagnostic sequence

Question	Signal to investigate
When do customers cancel?	Time from order placement to cancellation; compare against orders still at risk.
At what order stage?	Preparing, awaiting delivery partner, assigned, pickup delayed, en route.
Where and when?	Hourly demand, geography, restaurant clusters, peak-hour cohorts.
Why are orders unassigned?	Active courier supply, busy share, rejection rates, assignment latency, dispatch health.

Illustrative working scenario

Assume most problematic unassigned orders occur from 8-10 PM, roughly 90% of delivery partners are occupied, and dispatch logic is functioning normally. This points toward a capacity mismatch rather than a software bug. These are hypotheses for the exercise, not verified findings.

Root cause vs. symptoms

Likely underlying issue: peak-hour courier supply is insufficient for demand. Customer-facing symptoms: long assignment times, missed promises, uncertain ETA, cancellation risk. Coupons can help recover trust, but cannot add delivery capacity.

02 / Compare interventions

Option	Potential benefit	Tradeoff
Improve ETA and proactive alerts	Lower uncertainty; scalable baseline experience	Does not speed up delivery
Targeted recovery coupon	May prevent cancellation and encourage return	Discount leakage; margin pressure
Peak-hour courier incentives	Addresses available capacity	Cost; operational coordination
Batch pickup / route consolidation	Potentially fewer pickup trips	Routing complexity; freshness and ETA risk

Recommended product scope

Build the minimum instrumentation and customer communication needed to identify credible delays. Run a limited, randomized coupon experiment rather than offering compensation to all delayed orders. Explore supply-side fixes separately with operations.

Illustrative MVP behavior

- Store original promised ETA and updated predicted ETA; trigger only on meaningful predicted lateness.
- Notify affected customers early with an accurate revised ETA and normal support/cancellation options.
- Randomize eligible delayed orders into control, notification-only, and notification-plus-coupon arms.
- Offer a clearly described time-limited coupon to treatment users; record issuance, redemption, and cost.

Targeting hypothesis

Begin with delay severity and customer cancellation risk; evaluate recent order frequency and contribution economics as possible segments. Avoid assuming that high spend always means high profit or that loyal customers need the largest incentives.

03 / Experiment and measurement

Group	Experience	What it isolates
A: Control	Existing tracking experience	Current baseline
B: Transparency	Proactive revised ETA	Value of timely communication
C: Recovery	Revised ETA plus eligible coupon	Incremental effect of compensation

Success metrics

- Primary: cancellation rate among eligible delayed orders (predefine the eligibility rule).
- Secondary: 7- and 30-day repeat purchase, complaints per 1,000 orders, coupon redemption, support contacts.
- Business guardrail: incremental contribution margin net of incentive and operational costs.
- Quality guardrails: ETA accuracy, on-time rate, customer fairness, and refund/cancellation experience.

Illustrative pilot readout - not actual results

Metric	Notification only	Notification + coupon
Cancellation rate	14%	11%
30-day repeat purchase	32%	34%
Avg. incentive cost per eligible order	Rs. 0	Rs. 18
Complaints per 1,000 eligible orders	40	30

Interpretation

The coupon arm improves the cancellation rate by 3 percentage points relative to notification alone. That is three additional saved orders per 100 eligible orders. However, it also adds costs to the full eligible treatment group. A better cancellation metric does not by itself prove a better business outcome.

04 / Unit economics and decision

Illustrative one-period scenario

Assumption	Value
Eligible delayed orders	10,000
Incremental cancellation reduction vs. notification	3 percentage points
Additional completed orders	300
Contribution margin per additional completed order	Rs. 100
Avg. coupon cost per eligible order	Rs. 18

Incremental contribution from saved orders: $300 \times \text{Rs. } 100 = \text{Rs. } 30,000$.

Coupon expense: $10,000 \times \text{Rs. } 18 = \text{Rs. } 180,000$.

Net effect before incremental repeat-order value: $-\text{Rs. } 150,000$.

Break-even logic

With these inputs, each additional saved order effectively requires Rs. 600 in coupon spend ($\text{Rs. } 180,000 / 300$). If the margin from that saved order is only Rs. 100, the remaining Rs. 500 must be justified by truly incremental downstream value or lower costs. Repeat purchases should be evaluated against a randomized control, not inferred from coupon redemption alone.

Decision recommendation

Do not roll out a blanket Rs. 50 coupon based on this scenario. Keep proactive ETA communication as a baseline candidate, and restrict coupons to a testable, high-incrementality segment if the data supports it. Set an explicit margin hurdle before scaling.

What would change my mind?

- A reliable estimate showing incremental 30-day or 90-day contribution exceeds incentive cost.
- A narrower segment with materially higher cancellation reduction or lower coupon redemption cost.
- Evidence that non-monetary recovery is insufficient for severe delays.

05 / Product judgment and next steps

Key tradeoffs

- Trust vs. incentives: proactive honesty helps users, but compensation can be appropriate after serious failures.
- Retention vs. margin: coupon redemption is not the same as incremental customer value.
- Short-term vs. structural: service recovery cannot replace courier-capacity planning.
- Targeting vs. fairness: promotional offers may vary, while clear information and appropriate support should remain broadly accessible.

If this were a live PM assignment

- Validate data availability and baseline metrics with analytics and operations.
- Interview support/operations stakeholders and inspect real cancellation reasons.
- Run an appropriately powered randomized experiment with pre-agreed success and stop criteria.
- Investigate courier supply forecasting, shift incentives, and restaurant wait times in parallel.

Portfolio takeaway

The core lesson is not "coupons reduce cancellations." It is that a PM should diagnose the bottleneck, distinguish symptom management from root-cause work, and use experiments and unit economics to decide whether an attractive feature deserves investment.

Honest project status

Independent case exercise; no access to platform internal data, no production implementation, and no verified customer impact. All figures are illustrative. The deliverable is the diagnostic framework, prioritization, experiment design, and financial decision model.

Suggested interview summary

"I explored a food-delivery cancellation problem by tracing the order lifecycle and testing a peak-hour supply-shortage hypothesis. I compared operational and customer-facing solutions, then modeled a targeted compensation experiment. The illustrative economics challenged my initial preference for coupons, so I recommended a controlled pilot rather than a broad launch."